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Skill Builder : 2

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In [1]: cifar_h, cifar_w = 32, 32
        fashion_h, fashion_w = 28, 28

        print(f"CIFAR-10 Original: {cifar_h}x{cifar_w} → Resized: {cifar_h//2}x{cifar_w//2}")
        print(f"Fashion-MNIST Original: {fashion_h}x{fashion_w} → Resized: {fashion_h//2}x{fashion_w//2}")
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CIFAR-10 Original: 32×32 → Resized: 16×16
Fashion-MNIST Original: 28×28 → Resized: 14×14

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In [2]: import cv2
        import numpy as np
        import matplotlib.pyplot as plt

        data = np.load(r"C:\Users\HP\OneDrive\Desktop\CV&DL\student_micro_dataset2.npz")
        cifar_x = data['cifar'] # (50, 32, 32, 3)
        fashion_x = data['fashion'] # (50, 28, 28)

        # — CIFAR-10 —————
        fig, axes = plt.subplots(2, 5, figsize=(14, 5))
        fig.suptitle("CIFAR-10 | Top: Original (32×32) Bottom: Resized (16×16)", fontweight='bold')
        for i in range(5):
            original = cifar_x[i]
            resized = cv2.resize(original, (16, 16), interpolation=cv2.INTER_AREA)
            axes[0, i].imshow(original)
            axes[0, i].set_title(f"Orig {i+1}\n32×32")
            axes[0, i].axis('off')
            axes[1, i].imshow(resized)
            axes[1, i].set_title(f"Resized {i+1}\n16×16")
            axes[1, i].axis('off')
        plt.tight_layout()
        plt.show()

        # — Fashion-MNIST —————
        fig, axes = plt.subplots(2, 5, figsize=(14, 5))
        fig.suptitle("Fashion-MNIST | Top: Original (28×28) Bottom: Resized (14×14)", fontweight='bold')
        for i in range(5):
            original = fashion_x[i]
            resized = cv2.resize(original, (14, 14), interpolation=cv2.INTER_AREA)
            axes[0, i].imshow(original, cmap='gray')
            axes[0, i].set_title(f"Orig {i+1}\n28×28")
            axes[0, i].axis('off')
            axes[1, i].imshow(resized, cmap='gray')
            axes[1, i].set_title(f"Resized {i+1}\n14×14")
            axes[1, i].axis('off')
        plt.tight_layout()
        plt.show()
```

CIFAR-10 | Top: Original (32×32) Bottom: Resized (16×16)



Fashion-MNIST | Top: Original (28×28) Bottom: Resized (14×14)

